

ETHIOPIAN HEALTH PROFESSIONALS LICENSING EXAMINATION(EHPLE) PRACTICE TEST BOOKLET 2025

MEDICAL LABORATORY SCIENCE



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MINISTRY OF HEALTH-ETHIOPIA

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HEALTHIER CITIZENS FOR PROSPEROUS NATION!



Message from the State Minister, Ministry of Health – Ethiopia



Mrs. Frehiwot Abebe

Improving the quality of healthcare remains a critical global priority, essential for achieving sustainable development and universal health coverage. One of the most effective ways to ensure high standards in healthcare delivery is through the consistent assessment of professional competencies. In line with this commitment, the Ministry of Health of Ethiopia launched the Ethiopian Health Professionals Licensing Examination (EHPLE) in 2019 for undergraduates in seven core health disciplines. Since its inception, the program has expanded to encompass 17 health professions, reflecting our continued effort to build a competent and responsive health workforce.

The EHPLE was established with the primary aim of identifying and certifying health professionals who possess the minimum competencies required to provide safe and effective care. This ensures that all practicing professionals are equipped to meet the expectations of the communities they serve. The exam is coordinated by the Health and Health-Related Institutions and Professionals' Regulatory Lead Executive Office (LEO), which consists of four specialized desks. The LEO plays a vital role in system strengthening, enabling the exam to be conducted with increased efficiency, organization, and quality assurance.

It is important to recognize that the EHPLE is not a conventional academic or recruitment examination. Rather, it is a rigorous, competency-based assessment designed to evaluate the practical skills and critical thinking abilities that are essential for professional practice. This *Practice Items Booklet* is part of our broader strategy to support examinees and educators in preparing for the licensing process. It offers a realistic overview of the exam structure, promotes a deeper understanding of its objectives, and fosters transparency and sustainability in implementation.

The development of this booklet has been made possible through the collaborative efforts of experts from higher education institutions, the Addis Ababa University Testing Center, and professional associations. Their insights and dedication have been instrumental in shaping this resource. I extend my heartfelt thanks to all contributors and to the Ministry of Health team for their tireless work and steadfast commitment to excellence in healthcare regulation.

We believe this booklet will serve as a valuable tool for candidates preparing for the examination and for educators guiding them toward success. Together, we are building a stronger, more competent health workforce for Ethiopia.

Introduction EHPLE

Welcome to the 2025 Edition of the Ethiopian Health Professionals Licensing Examination Practice Test Booklet.

This resource is designed to support candidates in their preparation for the national licensing examination, ensuring they are well-equipped with the knowledge and clinical reasoning skills required to deliver competent and ethical healthcare across Ethiopia.

In Ethiopia, the licensing examination serves as a pivotal tool in evaluating the competency of health professionals who have completed their academic training and are ready to transition into independent practice. The exam not only ensures public safety by verifying professional readiness but also upholds national and international standards in healthcare service delivery. As the healthcare system continues to expand and evolve, a strong, well-regulated professional workforce becomes even more essential.

This practice booklet contains 200 multiple-choice questions modelled after the structure, content, and format of the actual licensing examination. The questions span across core clinical domains essential for entry-level practice. This includes areas such as Patient care, leadership and management, ethics, and Health Promotion as outlined in the Information Booklet 2024.

The booklet is intended to:

- Familiarize candidates with the structure and cognitive level of the actual licensing examination.
- Offer opportunities for self-assessment and identification of strengths and gaps in knowledge.
- Encourage critical thinking and clinical decision-making in scenarios reflective of real-world healthcare practice in Ethiopia.
- Reduce exam-related anxiety by providing realistic practice experiences.

Administered under the auspices of the Ministry of Health and relevant regulatory bodies, the Ethiopian Health Professionals Licensing Examination is grounded in evidence-based frameworks and aligned with both national needs and global best practices. It has been structured to assess a range of competencies, including medical knowledge, communication, professionalism, and safe clinical practice.

This practice test is not a predictive tool, and performance on it does not guarantee similar outcomes on the actual exam. However, it provides an invaluable opportunity for familiarization with question styles, pacing strategies, and content review. We strongly recommend that candidates simulate real testing conditions while using this resource, including timing their responses and reviewing rationale for both correct and incorrect answers.

The goal of licensing examinations is not only to regulate entry into the profession but also to inspire excellence and lifelong learning. By engaging with this practice test, you are taking a proactive step toward professional development, accountability, and a commitment to serving your patients with competence and compassion.

We wish you success on your licensing journey—and beyond.

Acknowledgements

This *Practice Test Items Booklet for the Ethiopian Health Professionals Licensing Examination* is the result of a collaborative effort among dedicated professionals committed to advancing the quality of health sciences education in Ethiopia. The Ministry of Health extends its sincere appreciation to all those who contributed their time, expertise, and insights to the development of this essential resource.

Special recognition is given to the item developers and reviewers from various higher education institutions—both public and private—whose academic rigor and practical experience were instrumental in shaping the content and ensuring its relevance to real-world healthcare practice. Their contributions have helped create a high-quality, standardized practice tool that reflects national competencies and global benchmarks.

We are especially grateful to the team at the Addis Ababa University Testing Center, whose technical support and academic guidance were critical throughout the process. Their leadership in assessment design and educational evaluation has significantly strengthened the quality of this booklet.

We also thank the committed staff of the Ministry of Health, particularly those within the Health and Health-Related Institutions and Professionals' Regulatory Lead Executive Office (HHRIPR LEOs), whose coordination, oversight, and unwavering dedication ensured the successful completion of this project.

Together, these collective efforts reflect our shared vision of building a competent, ethical, and resilient health workforce for Ethiopia.

Medical Laboratory Science

1. A Medical laboratory Scientist was requested to measure blood glucose concentration for a follow-up diabetic patient. To avoid glycolysis, which can decrease glucose concentration to 10 mg/dL per hour, the scientist needed to use the most anti-glycolytic agent to preserve glucose for up to three days and inhibit bacterial growth.

Which anti-glycolytic agent is most appropriate in this scenario?

- A. Sodium citrate
- B. Sodium heparin
- C. Sodium fluoride
- D. Sodium chloride

ANSWER: C

2. As part of quality assurance in clinical chemistry, a medical laboratory professional should minimize or control factors affecting results produced from the laboratory.

What is the most likely difficult factor to control by the medical laboratory professional?

- A. Posture
- B. Exercise
- C. Travel
- D. Altitude

ANSWER: D

3. By reviewing the workload for a day, a medical laboratory scientist working in the clinical chemistry section. The laboratory professional wants to prepare 2000 micro litter working solution to measure urea level. For preparing a working solution from urea stock solution, there is a formula stated as 2ml R1 + 3 ml R2.

What is the most appropriate amount of R1 that should be taken by the scientist?

- A. 750 micro litter
- B. 800 micro litter
- C. 1200 micro litter
- D. 1000 micro litter

ANSWER: B

4. A physician requested a medical laboratory scientist to measure the level of alanine transaminase for a patient suspected of heart disease. A laboratory professional measured these enzymes using a semi-automated machine and found an average absorbance of 0.6/120seconds. The factor for this enzyme is 1500 IU/L.

What is the most likely concentration of alanine transaminase?

- A. 7.5 IU/L.
- B. 450 IU/L.
- C. 350 IU/L.
- D. 10.5 IU/L.

ANSWER: B

5. A Medical Laboratory Professional working in a clinical chemistry laboratory has taken two absorbance measurements at 60 and 120 seconds consecutively to measure the concentration of serum urea.

What is the most likely assay technique used by the professional?

- A. Endpoint
- B. Kinetic
- C. Fixed time
- D. Bichromatic

ANSWER: C

6. A 26-year-old patient was brought to an emergency department suspected for diabetes mellitus. After blood collection, a laboratory analyst detected immunological markers that indicate cell-mediated autoimmune destruction of the Beta-cells of the pancreas. What is the most likely type of diabetes mellitus the patient was diagnosed with in this scenario?
- A. Type 1 diabetes
 - B. Type 2 diabetes
 - C. Mitochondrial diabetes
 - D. Gestational diabetes mellitus
- ANSWER: A
7. Blood glucose determination is carried out by different test methods, having different diagnostic specificity and sensitivity. For the patient who requested fasting blood glucose determination, a chemistry analyst needs to perform it with a gold standard test method to accurately identify the blood glucose level. Which is the most likely test method used in the scenario?
- A. Peroxidase
 - B. Hexokinase
 - C. Glucose oxidase
 - D. Glucose dehydrogenase
- ANSWER: B
8. A Medical laboratory scientist has collected specimens from a patient suspected for a cardiovascular disease risk factor and measures lipoprotein level using Polyanion (heparin) to separate any of the lipoproteins so that he will measure specific lipoproteins. What is the most likely principle used in this case?
- A. Electrophoretic
 - B. Chromatographic
 - C. Ultracentrifugation
 - D. Chemical precipitation
- ANSWER: D
9. A laboratory technologist measured serum alkaline Picric acid to generate an orange-red product. Fixed time assay was used to measure the rate of change in absorbance, and was measured by a spectrophotometer. What is the most likely specific marker measured by this principle?
- A. Creatinase
 - B. Creatinine
 - C. Creatine Kinase
 - D. Creatinine deaminase
- ANSWER: B
10. A 55-year-old man was admitted for severe liver disease; the blood sample was collected and measured for both Hepatic and Renal Function Tests. A clinician requested the tests interested to draw a conclusion about Renal Function tests before the hepatic tests. Which is the most likely reduced Renal Function test for this patient?
- A. Urea
 - B. Glucose
 - C. Creatinine
 - D. Cystatin C
- ANSWER: A

11. A Medical Laboratory Scientist working in a clinical chemistry laboratory is requested to determine the creatinine clearance value for a patient suspected of chronic kidney disease (CKD). The laboratory results revealed Creatinine 2 mg/dl, 24-hour urine volume 2000ml/24hrs, and urine creatinine 100 mg/dl.
What is the most likely value of creatinine clearance in this case?
- A. 64.4 ml/min
 - B. 67 ml/min
 - C. 54 ml/min
 - D. 60.5 ml/min
- ANSWER: A
12. A creatinine can be measured enzymatically and chemically (Jaffe Method), and there are interfering substances that interfere in both cases. Enzymatic and Jaffe test methods for Creatinine measurement can cause a negative bias in both test methods.
What is the most likely interfering substance in the case?
- A. Glucose
 - B. Uric acid
 - C. Bilirubin
 - D. Ascorbate
- ANSWER: C
13. A coupled enzymatic reaction that uses lactate dehydrogenase as an indicator of enzyme to catalyze the reduction of the formed pyruvate to lactate with simultaneous oxidation of NADH was used by a laboratory technologist. Then, the change in absorbance at 340nm was measured continuously.
What is the most likely enzyme determination in serum used in this principle?
- A. Lactate dehydrogenase
 - B. Alanine aminotransferase
 - C. Aspartate aminotransferase
 - D. Gamma glutamyltransferase
- ANSWER: B
14. A laboratory technologist performed a serum analysis and found that total proteins are within the acceptable normal range. However, the measurement for Albumin showed a very low level. Total bilirubin was slightly increased.
What is the most likely interpretation for this scenario?
- A. Early liver cirrhosis
 - B. Increased proteinuria
 - C. Protein-losing enteropathy
 - D. Decreased levels of globulins
- ANSWER: A
15. An immunoassay using chemiluminescent reagent that is linked to a second antibody to allow quantification is used by a laboratory technologist for the possible detection of cardiac disease.
What is the most likely specific marker measured by this principle?
- A. Troponin
 - B. Creatine kinase
 - C. Lactate dehydrogenase
 - D. Aspartate aminotransferase
- ANSWER: A

16. A medical laboratory professional working in a clinical chemistry department wants to measure enzyme activity; however, he observed that the reagent does not have a buffer solution. What is the most likely factor affecting the enzyme activity measurement?
- A. PH
 - B. Temperature
 - C. Enzyme concentration
 - D. Substrate concentration
- ANSWER: A
17. Thyroid function tests were measured in a clinical chemistry laboratory, and the results were normal for free thyroxine (FT4) and high for thyroid-stimulating hormone (TSH) level. Which is the most likely diagnosis in this scenario?
- A. Subclinical hypothyroidism
 - B. Primary hypothyroidism
 - C. Subclinical hyperthyroidism
 - D. Secondary hypothyroidism
- ANSWER: A
18. A patient with symptoms of loss of secondary sexual characteristics, who has hypothalamic dysfunction, as a result of reduced hypothalamic pulse generator frequency, came to a hospital. The levels of fertility panel were measured. A laboratory result indicated low testosterone levels, low FSH, and/or LH levels. What is the most likely diagnosis in this case?
- A. Primary Hypogonadism
 - B. Tertiary Hypogonadism
 - C. Secondary Hypogonadism
 - D. Hypergonadotropic-Hypogonadism
- ANSWER: B
19. A specimen was collected from a 30-year-old woman for monitoring trophoblastic tumour, and a Medical Laboratory scientist working in clinical chemistry suspected a falsely increased level of β -human chorionic gonadotropin. What is the most likely source of error for this tumour marker analysis?
- A. Cirrhosis
 - B. Pancreatitis
 - C. Pregnancy
 - D. Viral hepatitis
- ANSWER: C
20. A 55-year-old male outpatient has given a laboratory request paper to a receptionist for a potassium electrolyte test. ISEs consist of or are covered by a unique material that is more selective for one ion than other ions. Thus, a laboratory analyst should choose the selective membrane so as to selectively bind with potassium ions. Which is the most likely membrane used based on the above scenario?
- A. Valinomycin
 - B. Calcium oxide
 - C. Silicon dioxide
 - D. Lithium aluminium
- ANSWER: A

21. A virologist wants to grow a virus for clinical isolation. He used a cell derived from cancerous tissue and capable of serial cultivation indefinitely without senescing.
What is the most likely type of cell line is used by the virologist in this scenario?
- A. Continuous cell lines
 - B. Diploid cell strains
 - C. Primary cell culture
 - D. Human amnion cell culture
- ANSWER: A
22. A physician suspected a male patient for hepatitis B infection and sent him to a laboratory for serological tests. A phlebotomist collects the blood and runs serological tests for HBV. The patient was positive for IgM anti-HBc.
What is the most likely indication of this test result?
- A. Recent infection
 - B. Chronic infection
 - C. Past infection
 - D. Recovery
- ANSWER: A
23. A double-stranded RNA virus, which has 11 segments, was identified from a clinical specimen. The presumptive identification showed to be Rotavirus.
What could be the most prominent diseases caused by this isolate?
- A. Respiratory infection in infants and children
 - B. Respiratory infection in adults
 - C. Diarrheal illness in infants and children
 - D. Diarrheal illness in adults
- ANSWER: C
24. An infant was brought to a paediatric OPD health facility with destruction of motor neurons in the spinal cord, which resulted in flaccid paralysis.
What is the most likely disease observed in this scenario?
- A. HGV
 - B. Rubella virus
 - C. Rhinoviruses
 - D. Polio virus
- ANSWER: D
25. A man who had an illness characterized by hallucinations, combativeness, muscle spasms, signs of meningeal irritation, seizures, and focal paralysis was hospitalized. A physician diagnosed the patient and informed the parents that the chance of survival of the patient is almost zero.
What is the most likely causative agent for this disease?
- A. Parainfluenza virus type 1
 - B. Measles virus
 - C. Respiratory syncytial virus
 - D. Rabies virus
- ANSWER: D

26. A physician wanted to know the prognosis of a patient and monitor the effectiveness of antiviral therapies at the ART clinic. After assessing all clinical variables, he wanted to confirm it with important laboratory findings.
Which is the most likely type of test recommended for this purpose?
- A. HIV viral load
 - B. Pro virus PCR
 - C. HIV genotyping
 - D. HIV antibody test
- ANSWER: A
27. A laboratory technologist, who is working in a diagnostic mycology laboratory, received a specimen with a request for KOH for fungal diagnosis. The technologist identified the case of a specimen and treated it with 10 percent KOH.
What is the most appropriate next step in this case?
- A. Indian ink preparation
 - B. Stain with special fungal stain
 - C. Inoculation of the specimen into a fungal culture
 - D. Microscopic examination for fungi by light microscope
- ANSWER: D
28. Black Piedra is a fungal infection of the scalp hair, less commonly of the beard or moustache, and rarely of axillary or pubic hairs. The disease is characterized by the presence of discrete, hard, gritty, dark brown to black nodules adhering firmly to the hair shaft.
What is the most appropriate specimen needed to identify the causative agent in this case?
- A. Hair fragments
 - B. Skin scraping
 - C. Slit-skin
 - D. Punch biopsy
- ANSWER: A
29. In a kindergarten, where children share a towel and a bed, majority of them developed erythema, induration, itching, and scaling on their bodies. A kindergarten nurse, who cares for these children, was consulting physician, and she suspected cutaneous mycosis.
What is the most UNLIKELY agent for this scenario?
- A. Trichosporon
 - B. Epidermophyton
 - C. Malassezia
 - D. Hortaea (Exophiala)
- ANSWER: B
30. A male gardener, who is frequently working with soil and plants, came to a hospital with a complaint of skin lesion on one of his arms. He remembered that he had skin prick accident six weeks ago while he was planting rose flowers. At the beginning, it did not look bad; unfortunately, he had a serious infection with the development of multiple ulcers. Specimens collected from the skin lesion, stained and examined with a microscope, fungal elements, small spherical and elongated budding yeast cells was identified, which was an indicative of fungal infection.
What is the most likely causative agent in this case?
- A. *Rhinosporidiumseeberi*
 - B. *Wangielladermatitidis*
 - C. *Histoplasma capsulatum*
 - D. *Sporothrixschenckii*
- ANSWER: D

31. A female mycologist, who was working in a fungal diseases research institute, planned to work on subcutaneous mycosis etiological agents. She started working on the various fungal agents or genera associated with the particular group of diseases.

What is the most likely feature of the fungal etiologic agent in this group?

- A. It produces melanin
- B. It exists as a sole parasite
- C. Animals serve as reservoirs
- D. Limited geographic distribution

ANSWER: A

32. Skin scraping and nail clippings were collected from a patient in the ward and submitted to a mycology laboratory for dermatophyte culture. A technologist confirmed if the container was sterile, the specimen fulfilled all the criteria for specimen reception; and then, the technologist inoculated the specimen on mycosel agar.

What is the most likely minimum incubation period and optimal temperature respectively to report it as a negative?

- A. 7 days at 30 degrees centigrade
- B. 7 days at 37 degrees centigrade
- C. 21 days at 30 degrees centigrade
- D. 21 days at 37 degrees centigrade

ANSWER: C

33. A new male graduate laboratory technologist, who was assigned to a bacteriology laboratory, requested to prepare AFB stain reagent. He started preparing carbol fuchsin by mixing basic fuchsin with ethanol.

What is the most likely next solution to be used in this case?

- A. 95 percent absolute methanol
- B. 5 percent Phenol aqueous solution
- C. 3 percent acid alcohol
- D. 0.01M Potassium hydroxide

ANSWER: B

34. In a bacteriology laboratory, a microbiologist use 0.5 percent McFarland solution to standardize the inoculum amount., A microbiologist dissolves in the normal saline to perform an antibiotics susceptibility test.

What is the most appropriate storage condition for this solution?

- A. At room temperature on the table
- B. At a dark place in a sealed container
- C. In the refrigerator
- D. In deep freezer

ANSWER: B

35. A physician sends a wound swab to a bacteriology laboratory for Gram stain. A laboratory technologist prepared a smear, performed a gram stain appropriately, and reported the result as gram-positive.

What is the most likely colour of staining dye that the bacterium retains?

- A. Crystal violet
- B. Lugol's Iodine
- C. Safranin
- D. Neutral red

ANSWER: A

36. A patient suspected for tuberculosis came to a bacteriology laboratory with a request form for AFB microscopy. On the request paper, all the needed information was well written. A laboratory assistant instructed the patient to collect the best quality specimen to enhance the diagnosis. What is the most appropriate time to collect the specimen?
- A. Mid-night
 - B. Mid-day
 - C. Any time
 - D. Early morning
- ANSWER: D
37. A laboratory personnel, who is working in a health centre's laboratory, collect urine sample from a woman suspected for a urinary tract infection for urine culture to regional laboratory. It takes more than two hours to deliver the sample to the regional laboratory. What is the most appropriate recommended means to transport a urine sample?
- A. Transport it in a normal urine container
 - B. Use the cold chain to transport it
 - C. Add boric acid as a preservative
 - D. Add formaldehyde as a preservative
- ANSWER: C
38. A hospital laboratory safety officer orients a laboratory cleaner to decontaminate every culture plate and leftover of bacteriological specimen/culture before washing or disposing since they are infectious. What is the most appropriate way of decontamination in this case?
- A. Soak them in 70 percent alcohol
 - B. Autoclave them at 121 degrees centigrade for 15 minutes
 - C. Soak them in 5 percent sodium hypochlorite for 10 minutes
 - D. Soak them in absolute alcohol
- ANSWER: B
39. A male laboratory assistant, who works in a bacteriology laboratory, prepared different types of culture plates for research purposes. He labelled each plate with batch numbers, date, and stores. What is the most likely recommended storage condition for this prepared culture plate?
- A. On the shelf at room temperature
 - B. In the biological safety cabinet
 - C. At 2-8 degrees centigrade in the sealed plastic bag
 - D. In the incubators at room temperature
- ANSWER: C
40. A microbiologist, who wants to perform an antibiotic susceptibility test for the gram-negative bacteria, which were isolated from a patient sample by using the Kirby-Bauer disc diffusion technique. What is the most likely recommended culture media in this case?
- A. MacConkey agar
 - B. Nutrient agar
 - C. Blood agar
 - D. Mueller Hinton agar
- ANSWER: D

41. A male graduate student of medical laboratory science, who is working on his thesis on antibiotic susceptibility patterns, performed a disk diffusion test on the bacteria isolated from wound swab. He wants to interpret the inhibition zone of each antibiotic. What is the most likely recommended guideline to interpret antibiotic inhibition zone?
- A. National treatment guidelines
 - B. Manufacturer leaflets
 - C. NCCLS guidelines
 - D. Institutional guidelines
- ANSWER: C
42. A swab sample collected from a surgical site with signs of infection was inoculated on blood agar and nutrient agar. On blood agar, the isolates show a clear zone of hemolysis (beta haemolysis). On nutrient agar, the colonies are golden yellow. The isolates were coagulase and catalase-positive. What is the most likely isolated bacterium in this case?
- A. *Staphylococcus aureus*
 - B. *Streptococcus agalactiae*
 - C. *Streptococcus pyogenes*
 - D. *Staphylococcus lugdunensis*
- ANSWER: A
43. A woman, who was suffering from a toothache, went to a nearby dental clinic for treatment. Upon examination, a physician observed that the cavity formation in the tooth is most probably due to the destruction of its mineralized tissues. A Gram stain result showed Gram-positive cocci arranged in long chains. A physician told her that she had dental caries due to bacterial infection. What is the most likely bacterial etiology in this case?
- A. *Streptococcus pneumoniae*
 - B. Viridans streptococcus
 - C. *Staphylococcus saprophyticus*
 - D. *Staphylococcus epidermidis*
- ANSWER: B
44. A wound swab collected from an admitted patient at a surgical ward was received for culture and antibiotic sensitivity testing. It was inoculated in different culture media and incubated at 37 degrees centigrade for 24 hours. *Streptococcus species* was identified by using different biochemical tests. A technologist wants to continue the antibiotic sensitivity testing and wants to choose among the available culture media. What is the most appropriate media to conduct antibiotic susceptibility test in this case?
- A. Blood agar media
 - B. Muller Hinton media
 - C. Chocolate media
 - D. Nutrient media
- ANSWER: A
45. A 10-year-old child came to a hospital with signs of oedema of the tonsils, uvula, submandibular region, and anterior neck (bull neck). A physician suspected that the child was infected with *Corynebacterium diphtheria*. What is the most likely virulence factor of this bacterium?
- A. Endotoxin
 - B. Capsule
 - C. Spore
 - D. Exotoxin
- ANSWER: D

46. A farmer came to a hospital with a spreading oedematous myonecrotic wound, with which extravasation of blood reduces and blood supply to the affected part, and the person developed gas gangrene. A physician suspected; it is secondary to bacterial infection. What is the most likely bacterium that causes this disease?
- A. *Bacillus anthrax*
 - B. *Clostridium perfringens*
 - C. *Bacillus cereus*
 - D. *Clostridium tetani*
- ANSWER: B
47. A boy came to a nearby hospital after he got cut with a rusted knife while he was working in the garden with complaints of muscle stiffness, wound infection, and locked jaws. What is the most likely bacterium that causes this infection?
- A. *Clostridium difficile*
 - B. *Clostridium botulinum*
 - C. *Clostridium tetani*
 - D. *Bacillus anthrax*
- ANSWER: C
48. A laboratory technologist working in a bacteriology laboratory inoculates a urethral swab on blood and chocolate agar media and incubates aerobically at 37 degree centigrade for 24 hours. Gram stain was done from the culture, and Gram-negative diplococcic is seen in pairs. The technologist confirmed *Neisseria spp* grown and he wants species identification at least to differentiate between *Neisseria gonorrhoeae* from *Neisseria meningitides* by their carbohydrate fermentation profile. What is the most likely carbohydrate used to differentiate the species?
- A. Glucose
 - B. Maltose
 - C. Lactose
 - D. Sucrose
- ANSWER: B
49. A stool sample collected from a patient with fever, headache, and abdominal pain was sent to a bacteriology laboratory for stool culture. The stool sample was inoculated in xylose, lysine deoxycholate agar, and pink colonies with black centers were grown on a culture plate. What would be the most likely causative agent identified in this case?
- A. *Salmonella Typhi*
 - B. *Salmonella Paratyphi A*
 - C. *Salmonella Paratyphi B*
 - D. *Salmonella Paratyphi C*
- ANSWER: A
50. A urine sample collected from a woman, who was suffering from recurrent urinary tract infection was sent to a bacteriology laboratory for urine culture. The specimen was inoculated to MacConkey agar and blood agar plates and incubated at 37 degrees centigrade for 24 hours. Gram-negative lactose fermenter bacteria growth seen on MacConkey agar. What is the most likely bacterium identified from the specimen?
- A. *Proteus vulgaris*
 - B. *Escherichia coli*
 - C. *Salmonella Typhi*
 - D. *Pseudomonas aeruginosa*
- ANSWER: B

51. After heavy flood in a village, epidemic disease with characteristics of vomiting and watery diarrhoea occurred. Environmental health officer suspected it could be cholera and awared hospital laboratory to be ready for the diagnosis.
What is the most appropriate selective media the laboratory should prepare?
- A. MacConkey agar
 - B. TCBS agar
 - C. XLD agar
 - D. Chocolate agar
- ANSWER: B
52. A sputum sample collected from suspected individual with pulmonary tuberculosis was sent to a bacteriology laboratory for AFB examination. The result was found to be AFB positive and reported as AFB.
What is the most likely morphological feature of the bacteria in this case?
- A. Red coccobacilli
 - B. Pink short rods
 - C. curved red rods
 - D. Red short rods
- ANSWER: C
53. A 15-year-old boy came to a hospital with complain of swelling of lymph node at the left parts of his neck. An attending physician suspected tubercular lymphadenitis and collected the sample for culture to identify *M. tuberculosis*.
What is the most likely media would be used in this case?
- A. Blood agar
 - B. MacConkey agar
 - C. Lowenstein-Jensen (LJ)
 - D. Mannitol salt agar (MSA)
- ANSWER: C
54. A 50-year-old woman came to a hospital with the sign of follicular conjunctivitis with diffuse inflammation on the entire conjunctiva. A physician diagnoses the patient and rules out trachoma.
What is the most likely bacterial agent causing this infection?
- A. *Rickettsia rickettsiae*
 - B. *Rickettsia prowazekii*
 - C. *Chlamydia trachomatis*
 - D. *Chlamydia psittaci*
- ANSWER: C
55. A laboratory technologist, who was working in a molecular biology laboratory, prepared the laboratory for the extraction of Nucleic acids from the blood sample. He has planned to extract and purify the nucleic acids in traditional methods by heating the sample in the presence of detergents and phenol solution. Furthermore, he has arranged the removed contaminants by equilibrium density gradient centrifugation in cesium chloride.
What is the most likely nucleic acid to be extracted as per the technologist's work plan?
- A. DNA
 - B. RNA
 - C. Cesium chloride
 - D. Difficult to extract
- ANSWER: A

56. A laboratory technologist, who was in a molecular biology laboratory, extracted nucleic acids from biological samples. Next, he assessed the purity of the extracted samples by taking spectrophotometer readings at 260 and 280 nm. Accordingly, the ratio of the extracted sample absorbance at 260/280 was 1.9.

What is the most likely absorbance ratio that indicates the nucleic acid status?

- A. Lipid contamination
- B. Protein contamination
- C. No contamination
- D. phenol contamination

ANSWER: C

57. A laboratory technologist, who got an order to determine the viral load of an HIV/AIDS patient, from the given biological sample. The technologist prefers to run RT-PCR (Reverse Transcription-PCR) procedure.

What is the special feature of RT-PCR that makes it the most appropriate choice in this case?

- A. Coverts DNA into mrna
- B. Converts RNA to DNA
- C. Converts RNA to cdna
- D. Coverts protein to DNA

ANSWER: C

58. A laboratory technologist, who was working in a given molecular biology laboratory, planned to perform Southern blotting techniques. Accordingly, after extracting and purifying DNA, he runs agarose gel electrophoresis. He also added a buffer to this agarose gel to separate the double-strand DNA, to facilitate blotting of single strands to the nitrocellulose paper.

What is the most likely buffer added to agarose gel in this case?

- A. Strong Acidic
- B. Weak acidic
- C. Strong alkaline
- D. Weak alkaline

ANSWER: C

59. Senior medical laboratory technology students had the opportunity to visit the molecular biology laboratory. During their visit, the laboratory head explained to them about the principles and purposes of RNA blotting methods for the students.

What is the most likely type of blotting explained in this case?

- A. Eastern
- B. Northern
- C. Southern
- D. Western

ANSWER: B

60. A laboratory technologist, who was working in a given molecular laboratory, usually undertook different actions to reduce contamination during PCR procedures.

What is the most likely action to be applied for improving PCR results?

- A. Working with bare hands in a 70 percent ethanol cleaned biosafety hood
- B. Use of aerosol-free pipette tips during PCR sample processing
- C. Process the PCR sample within the same time interval gaps
- D. Process pre-PCR and post-PCR solutions in the same room

ANSWER: B

61. A urine specimen with increased turbidity and foul odour was brought to a laboratory for investigation. The chemical analysis showed a negative dipstick measurement for all tests. What is the most likely explanation for this specimen?
- A. It is not a fresh specimen
 - B. It has been refrigerated
 - C. There is limited bacterial growth
 - D. It used chemical preservatives
- ANSWER: A
62. A urine specimen was collected for routine urinalysis. However, a laboratory professional rejected the urine specimen before further analysis. What is the most likely reason for a technologist to reject the specimen?
- A. Specimen contaminated with faeces
 - B. Matching labels and requisition form
 - C. Specimen preserved with chemicals
 - D. Specimens of sufficient quantity
- ANSWER: A
63. A patient with a history of high alcohol consumption visited a clinic. The screening tests for bilirubin were positive, and yellowing of the whites of the eye. The doctor requested a serum and urine bilirubin measurements to the nearby regional laboratory, and he was frustrated when the result for both serum and urine bilirubin measurements came back negative. What would be the most likely precautions missed while handling this specimen?
- A. The correct specimen was sent
 - B. The specimen was refrigerated
 - C. The specimen was exposed to light
 - D. Specimen was preserved with chemicals
- ANSWER: C
64. A Physician collected a specimen using the needle aspiration technique which is called arthrocentesis, and used to analyse fluid build-up in a patient. A syringe that was moistened with heparin was used to prevent fluid clotting. What is the most likely fluid analysis used in these materials?
- A. Serous fluid
 - B. Pleural fluid
 - C. Amniotic fluid
 - D. Synovial fluid
- ANSWER: D
65. A 34-year-old Rh-negative pregnant woman who had stillbirths was previously suspected of having dangers of anaemia due to Hemolytic Disease of the Newborn(HDN). Amniocentesis was performed, and amniotic fluid was sent to the laboratory for further chemical testing of bilirubin. However, separation of cellular elements and debris was not performed. What is the most likely source of error in processing this fluid without centrifugation?
- A. Increased cellular metabolism causes distortion of chemical constituents
 - B. Bilirubin measurement is not affected as long as it is protected from light
 - C. Repeated freeze-thawing for fetal lung maturity testing
 - D. Bilirubin measurement has an indirect(inverse) relationship with fetal anaemia
- ANSWER: A

66. A urine specimen with a fruity, sweet odour was brought to a laboratory. Without doing further tests, the laboratory scientist performed a specific reagent strip test. What specific strip test is the most likely positive for this specimen?
- A. Ketones
 - B. Bilirubin
 - C. Glucose
 - D. Protein
- ANSWER: A
67. A Patient was advised to have a clearance test performed. After the phlebotomy procedure is done, he then brought a 24-hour urine as instructed. However, the laboratory scientist spilled out small amount of urine that was collected and proceeded with clearance test calculations. What is the most likely erroneous result in the above scenario?
- A. Creatinine level
 - B. Urea level
 - C. Volume of urine
 - D. Inulin level
- ANSWER: C
68. A pregnant woman brought a urine specimen to a laboratory, and her urine is yellow in colour with a large amount of white foam when shaken. What is the most likely interpretation of this result?
- A. Protein
 - B. Chyluria
 - C. Urobilin
 - D. Urobilinogen
- ANSWER: A
69. A CSF specimen with increased protein and lipid concentration was brought to a laboratory. It also contains different WBCs. What is the most appropriate kind of reporting format that could be used for these appearances of CSF?
- A. Turbid
 - B. Oily
 - C. Clotted
 - D. Pellicle
- ANSWER: A
70. A physician performed a procedure to take synovial fluid from the joints. When the fluid reached the laboratory, a medical laboratory scientist observed an uneven distribution of blood in the specimen. What is the most likely reporting format that could be used for this specimen?
- A. Haemorrhagic arthritis
 - B. Traumatic aspiration
 - C. Contamination of the specimen during handling
 - D. It should not be reported
- ANSWER: B

71. A urine chemical analysis on the specimen that has been refrigerated is used by a laboratory scientist before it returns to room temperature. As a result, a report by a Medical Laboratory Scientist is found to be false negative.

What examination is most likely affected in this case?

- A. Protein
- B. Glucose
- C. Ketones
- D. Haemoglobin

ANSWER: B

72. A technologist performed a chemical analysis of urine by using the principle of 'error of indicators' that produces a visible colorimetric reaction. Certain indicators change colour of indicators at constant pH in the presence of this substance.

What is the most likely test that could be detected by this principle?

- A. Protein
- B. Bilirubin
- C. Haemoglobin
- D. Ketone

ANSWER: A

73. A pale-yellow urine specimen was brought to the laboratory department. A laboratory technologist left the specimen for 30 minutes on the bench before analysing it as it was lunchtime. Hence, the specimen was exposed to light.

What is the most likely test that could have a false negative reading in this scenario?

- A. Urobilinogen
- B. Protein
- C. Leukocyte esterase
- D. Urobilin

ANSWER: A

74. A Medical laboratory scientist, who was working at the urinalysis unit, collected a urine specimen and found a red coloured urine specimen. He wanted to differentiate between haemoglobinuria and haematuria.

What is the most likely method that could be used to differentiate the two specimens?

- A. Reagent strip tests
- B. Microscopic examination
- C. By observing the appearance of red colour
- D. By dissolving the RBCs with 2 percent acetic acid

ANSWER: B

75. A CSF specimen with a slight turbidity was brought to a laboratory. The chemical examination for protein analysis of the CFS specimen showed an elevated level.

What is the most likely interpretation of the above-mentioned results?

- A. Traumatic tapping
- B. CSF leakage
- C. Rapid CSF production
- D. Meningitis

ANSWER: D

76. A laboratory scientist performed chemical analysis of urine by using a reagent strip and reported all results within 30-60 seconds. Then, all results were read as negative. What is the most likely chemical test that could be affected by this report?
- A. Glucose
 - B. Nitrite
 - C. Leukocyte esterase
 - D. Urine protein level
- ANSWER: C
77. A laboratory scientist was performing a microscopic examination of urine by the conventional glass-slide method. He was mistakenly allowed the specimen to flow outside the cover slip, which might result in the loss of heavier elements. What is the most likely affected result in this scenario?
- A. Cast
 - B. Bacteria
 - C. RBC
 - D. WBC
- ANSWER: A
78. An alkaline fresh urine specimen with an increased turbidity and foul Odor was brought to a laboratory. Nitrite test was negative, whereas leukocyte esterase was trace during chemical analysis by using reagent strips. What is the most likely finding expected to be seen under the microscope?
- A. Bacteria
 - B. RBC
 - C. Epithelial cells
 - D. Urate crystals
- ANSWER: A
79. A senior Medical Laboratory Scientist wanted to examine urine microscopy at low-power magnification. He usually used a cell which has irregular cytoplasm and a prominent nucleus that could serve as a good reference for focusing of the microscope. What is the most likely finding in this case?
- A. Epithelial cells
 - B. White blood cells
 - C. Red blood cells
 - D. Mononuclear cells
- ANSWER: A
80. A laboratory scientist identified epithelial cells under the microscope. The epithelial cells observed have a spherical shape with a characteristic centrally placed nucleus, suspected of emerging from the lining of the renal pelvis and bladder. What would be the most likely type of epithelial cell identified based on the listed feature?
- A. Renal tubular
 - B. Transitional
 - C. Squamous
 - D. Hyaline
- ANSWER: B

81. A young woman came to a laboratory for a haematological test that requires a large volume of blood. A phlebotomist greeted her and requested that sit down comfortably. Then, he started assembling the materials for venous blood collection.
What are the most likely required materials for this case?
- A. Haematology analyser, vortex, micropipette, centrifuge
 - B. Pasteur pipette, lancet, differential counter, haemocytometer
 - C. Tourniquet, syringe, disinfectant, test tube
 - D. Sealant, capillary tube, Hemocue, microscope slide
- ANSWER: C
82. A clinician refers a patient suffering from multidrug-resistant tuberculosis infection to the laboratory for erythrocyte sedimentation rate (ESR). The laboratory personnel planned for venous blood collection to be performed using the Westergren technique.
What is the most likely anticoagulant to be used in this case?
- A. Disodium EDTA
 - B. Trisodium citrate
 - C. Sodium Oxalate
 - D. Heparin
- ANSWER: B
83. A laboratory technologist received multiple blood smears for annual competency testing from an external body. The laboratory technologist is expected to assess the features and the quality of the smears macroscopically as well as microscopically.
What is the most appropriate features should the technologist should consider for this scenario?
- A. Wide and covering the whole slide
 - B. Free from lines and holes
 - C. Has a jagged tail
 - D. Short
- ANSWER: B
84. A microbiologist was preparing a 10 percent Giemsa working solution in the laboratory. He started by measuring 45ml of buffered water and added it to a cylinder.
What is the most likely amount of Giemsa stock solution that has to be added to the cylinder?
- A. 50ml
 - B. 20ml
 - C. 10ml
 - D. 5ml
- ANSWER: D
85. A research assistant was observing the quality of the blood films for peripheral morphology. After staining with Wright stain at the optimum PH value, he used tap water to wash off the stains. During microscopic evaluation, he noticed that the blood films were too pink.
What could be the most likely source of error in this case?
- A. The slide is pink
 - B. The tap water is acidic
 - C. The staining rack is unclean
 - D. The staining solution is unfiltered
- ANSWER: B

86. A 4th year student was performing differential white cell count on an apparently healthy control study participant to determine the relative count. The blood film was well prepared and well stained using Romanowsky stains. He faced difficulty identifying Neutrophil from Eosinophilic and asked his supervisor.

What is the most likely key feature is NOT used in this case?

- A. Size
- B. Number of lobes
- C. Primary granules
- D. Specific granules

ANSWER: C

87. A female laboratory technician was preparing to perform erythrocyte sedimentation rate (ESR) analysis. She collected a blood sample from her client and proceeded to perform the test.

What is the most likely material needed to do this test?

- A. Sahili pipette
- B. Westergren rack
- C. Microscope slide
- D. Centrifuge

ANSWER: B

88. A laboratory personnel was reviewing the different methods used for haemoglobin measurement. Among the manual techniques, he had chosen to analyse the acid-haematin method.

What is the most likely disadvantage of this method?

- A. Expensive
- B. Inaccurate
- C. Long procedure
- D. Use a large volume of blood

ANSWER: B

89. A laboratory technologist was inspecting a blood film of anaemia suspected patient. After noticing significant polychromasia, he decided to stain the leftover blood sample using a supravital stain. Then, he inspected the morphology closely to identify and report the relative count of Reticulocyte.

What is the most likely characteristic to be considered?

- A. Reticulocytes have an irregular shape
- B. Mature RBCs are larger in size
- C. Reticulocytes have precipitates
- D. Reticulocytes have nucleus

ANSWER: C

90. A Medical laboratory technologist assigned in the haematology performed haemoglobin value, packed cell volume and RBC count manually for the client, and the result indicates 11g/dl, 35 percent and 4×10^6 per micro litre respectively.

What is the most likely Mean Cell Haemoglobin Concentration (MCHC) for this client?

- A. 27.5g/dl
- B. 31.4g/dl
- C. 36.4g/dl
- D. 11.4g/dl

ANSWER: B

91. A 71-year-old man visited the health centre after exhibiting symptoms of dizziness and fatigue. Complete blood cell analysis was performed using a haematology analyser. The result revealed that the mean cell haemoglobin concentration (MCHC) of the patient is 25g/dl (low). What is the most likely type of anaemia in this case?
- A. Microcytic Hypochromic
 - B. Normocytic Normochromic
 - C. Macrocytic normochromic
 - D. Microcytic normochromic
- ANSWER: A
92. A female research assistant was reviewing the blood film of a study participant. She observed poikilocytosis and 15 percent sickle cells. She decided to run another investigation by placing test and normal red cells in a series of graded-strength sodium chloride solutions, by comparing the haemolysis in each tube to a 100 percent standard. What is the most likely test that best corresponds with this principle of investigation?
- A. Erythrocyte sedimentation rate
 - B. Osmotic fragility test
 - C. Electrophoresis
 - D. Serum iron test
- ANSWER: B
93. A 35-year-old pregnant woman came to a health centre for a check-up. The complete blood cell count revealed a decreased value of RBC indices. Further morphology analysis revealed the size of the red cells was variable. What is the most likely description of the appearance of red cells?
- A. Poikilocytosis
 - B. Acanthocytosis
 - C. Anisocytosis
 - D. Echinocytosis
- ANSWER: C
94. A female laboratory scientist was reviewing a 51-year-old male patient's blood film post-splenectomy. She observed a few red cells with inclusions composed of nuclear remnants (DNA) that appeared as a dark purple spherical granule, usually near the periphery of the cell. What is the most likely erythrocyte inclusion observed in this case?
- A. Basophilic stippling
 - B. Pappenheimer bodies
 - C. Howell jolly bodies
 - D. Heinz body
- ANSWER: C
95. A laboratory scientist was running an extensive blood test on a patient. Morphological examination revealed that the patient has microcytic/hypochromic anaemia. In order to further specify the diagnosis, he performed a test that identifies haemoglobin defects based on differences in their electrical charges. What is the most likely test the laboratory scientist is performing in this case?
- A. Complete blood cell count
 - B. Direct Antibody test
 - C. Electrophoresis
 - D. Sahli/Hellige Method
- ANSWER: C

96. An internist was reviewing the blood test results of a trauma patient. The morphology examination indicated reticulocytosis and spherocytosis; CBC indicated the raised haemoglobin value; and an extended blood panel revealed an increase in serum level of unconjugated bilirubin and lactic dehydrogenase (LDH) while it showed a decrease in serum haptoglobin level.

What is the most likely interpretation of the laboratory report in this case?

- A. Haemolytic anaemia
- B. Iron deficiency anaemia
- C. Pernicious anaemia
- D. Pyruvate Kinase deficiency

ANSWER: A

97. A laboratory technologist collects a blood sample from a patient exhibiting splenomegaly. A consistent and highly elevated level of red blood cell count, Hgb, Hct and the indices are noticed. The current analysis of the CBC indicates a significant increase in the red cell count and other red cell-related parameters from the last analysis. The peripheral blood cell morphology indicates basophilia, normal platelets and microcytic/normochromic red cells.

What is the most likely disorder affecting this patient?

- A. Anaemia
- B. Cushing Syndrome
- C. Polycythaemia Vera
- D. Essential thrombocythemia

ANSWER: C

98. A student was examining the blood smear morphology of a fellow student with flu-like symptoms during haematology laboratory class. He encountered a high amount of reactive lymphocytes. He also performed a complete blood cell count, which shows a white cell count of 14,000/ micro litre.

What is the most likely condition for this student?

- A. Viral Infection
- B. Myeloproliferative disease
- C. Hodgkin's Lymphoma
- D. Multiple myeloma

ANSWER: A

99. A laboratory technologist was performing a coagulation test for a patient suffering from prolonged bleeding time. The international normalized ratio (INR) of this patient is 2.5.

What is the most likely primary clinical significance performed in this test?

- A. Treatment monitoring
- B. Rule out other disorders
- C. Confirm infectious agents
- D. Verify the validity of laboratory test results

ANSWER: A

100. A haematology graduate student was performing running coagulation profile tests for patients suffering from bleeding disorders. He measured the time required for bleeding to stop from small subcutaneous vessels that have been severed by a standardized incision.

What is the most likely test the student is performing in this case?

- A. Coagulation time
- B. Clotting time
- C. Prothrombin time
- D. Bleeding time

ANSWER: D

101. A 46-year-old patient was in a pathologist's exam room for bone marrow aspirate collection. A laboratory technician was gathering materials for bone marrow smear preparation. Which material is NOT most likely to be required for this test?
- A. Microscope slide
 - B. Pasteur pipette
 - C. Spreader slide
 - D. Plain glass tube
- ANSWER: D
102. A laboratory technologist was performing a complete blood cell count using an automated analyser. The analyser detected and measured changes in the resistance produced by cells as they crossed through a small aperture. What is the most likely principle of this analyser?
- A. Optical Light Scatter
 - B. Electrical impedance
 - C. Flowcytometry
 - D. Pattern recognition
- ANSWER: B
103. An Immunohematologist was preparing to perform ABO blood grouping for a volunteer donor. He mixed Anti-A, anti-B and anti-D antisera with a drop of the donor's blood cell and waited for 2 minutes to see the formation of agglutination. What is the most likely type of ABO blood grouping performed in this case?
- A. RhD grouping
 - B. Cross-matching
 - C. Reverse grouping
 - D. Forward grouping
- ANSWER: D
104. A 35-year-old female in the bone marrow transplant unit received an apheresis platelet transfusion. After ten minutes of the infusion, her blood pressure dropped and her temperature increased. The infusion was immediately discontinued, blood cultures were drawn, and the remaining platelet product was sent to the blood bank. Transfusion reaction investigation revealed no clerical discrepancy or evidence of haemolytic reaction. Gram stain of the product in the platelet components revealed Gram-negative rods. Cultures of the remaining platelet products grew *Escherichia coli*. What is the most likely reason for the above condition?
- A. Transfusion-related infection
 - B. Transfusion-related acute lung injury
 - C. Red blood cell antigen-antibody reaction
 - D. Transfusion-associated circulatory overload
- ANSWER: A
105. A patient receiving ceftriaxone developed haemolytic anaemia. The direct antiglobulin test (DAT) is positive (IgG +C3). The patient's serum reacts with all test RBCs (no drug added) by the indirect antiglobulin test, but an eluate prepared from the patient's RBCs is nonreactive. What is the most likely explanation of these results?
- A. idiopathic warm autoantibody
 - B. Non-immunologic protein adsorption
 - C. drug-independent antibody (e.g., due to fludarabine)
 - D. circulating drug (or drug-immune complexes) in the patient's sample
- ANSWER: D

106. Individuals who have received injections of human growth hormone are permanently deferred from blood donation because of the possible disease transmission.
Which type of disease is the most likely to be transmitted if such an individual donates blood?
A. AIDS
B. Hepatitis
C. infectious mononucleosis
D. CreutzfeldtJakob–disease
ANSWER: D
107. Haematologist was performing cross-matching for blood transfusion. He mixed patient serum and donor RBC suspension in a test tube and incubated at 37 °C. After decanting the serum completely, he washed the cells three times in saline, added Anti-human Globulin (AHG) and mixed. The result showed visible agglutination.
What is the most likely indicated by these findings in this case?
A. Compatibility
B. Incompatibility
C. Adverse transfusion reaction
D. Differ from issuing the blood
ANSWER: B
108. A 17-year-old man was admitted to a hospital following severe dehydration. His hematocrit and haemoglobin values were 60 percent (high) and 19.8 g/dl (high), respectively. He needs to have transfusion therapy.
Which blood component would be the most appropriate in this case?
A. Fresh frozen plasma
B. Packed red cell
C. Whole blood
D. Plasma
ANSWER: D
109. A 4th year Medical laboratory student wanted to know his blood group. He started to collect the material need to do the test. He puts the following materials on the bench: a clean dry microscope slide, anti-A Antisera and anti-B Antisera, Capillary blood, a toothpick for mixing, a Lancet, alcohol 70 percent, cotton and a pencil. However, he realised that one material was missing from the list.
What is the most likely material missed in this case?
A. Syringe
B. Anti-ASO
C. Anti-D serum
D. Anti-MHC serum
ANSWER: C
110. The serum containing antibodies was incubated with erythrocytes containing antigens that adsorb the incomplete antibodies. After washing to dilute the excess antibody in the serum, addition of anti-globulin serum produces agglutination in the presence of incomplete antibodies.
What is the most likely test principle is indicated in this case?
A. Reverse grouping
B. Direct Coombs' Test
C. Compatibility Testing
D. Indirect Coombs' Test
ANSWER: D

111. A Medical laboratory technologist wanted to prepare a volume (V) of 100 ml of 10 percent concentration (C) working solution from a 95 percent stock solution.
What is the most likely method used in this case?
- A. Serial dilution
 - B. The dilution factor method
 - C. The C_1V_1 is equal to the C_2V_2 method
 - D. No possibility to do such working solution
- ANSWER: C
112. A technologist working at a primary hospital received an order from a physician to perform an ASO test. He collected a blood sample from the patient, processed the sample and did an ASO test following standard operating procedure. After a few minutes, he observed agglutination formed and the negative control test was negative.
What would be the most likely interpretation of this test result?
- A. Invalid
 - B. Positive
 - C. Negative
 - D. Wrong procedure
- ANSWER: B
113. A 40-year-old man visited his physician with complaints of generalised lymphadenopathy, or enlargement of the lymph nodes, malaise, fever, pharyngitis, and rash on the skin and mucous membranes. A clinician ordered the laboratory to do a Rapid plasma reagin (RPR) test.
What is the most likely clinical application of this test?
- A. Screening of Syphilis
 - B. Screening for Streptococcus pyogenes
 - C. Semi-quantitative determination of HCG
 - D. Determining the antecedent Group A streptococcus infection
- ANSWER: A
114. A 50-year-old female patient came to a clinic with complaints of morning stiffness around the joints lasting at least 1 hour, swelling of the proximal interphalangeal and wrist joints. After physical examination, the doctor ordered RF test, and it became positive.
What is the most likely clinical application of this test?
- A. Rheumatoid Arthritis
 - B. Multiple sclerosis (MS)
 - C. Myasthenia gravis (MG)
 - D. Autoimmune thyroid diseases (AITDs)
- ANSWER: A
115. A 60-year-old female visited the nearby medical centre with complaints of diffuse symmetrical joint pain, swelling, and stiffness of peripheral joints. After physical examination, the doctor ordered RF test. The laboratory technologist on duty mixed latex particles coated with human gammaglobulin with the Patient's sample, but no visible agglutination formed. Note: both control results had acceptable.
What is most likely indicated by the test result in this case?
- A. Invalid
 - B. Positive
 - C. Negative
 - D. Wrong procedure
- ANSWER: C

116. A 25-year-old woman had missed her menstrual period three weeks ago. She suspected pregnancy and she went to her primary care provider for confirmation. What is the most likely common test in this case?
- A. HCG
 - B. ANA
 - C. HVB
 - D. ELISA
- ANSWER: A
117. A young woman recently discovered that her boyfriend has confirmed HIV infection. She was concerned that she may have also contracted the infection, because she had experienced flu-like symptoms a month ago. She decided to visit her physician for a medical evaluation, but her test result became negative. Then, he prescribed molecular analysis, and the result was positive. What is the most likely interpretation of the first-round test result?
- A. True negative
 - B. True positive
 - C. False positive
 - D. False negative
- ANSWER: D
118. A healthy newborn developed a fever and a respiratory infection a few days after birth. He also has a slightly enlarged liver. The newborn's family has a cat that spends a great deal of time outdoors. The newborn's mother cleans the cat's litter box and does so throughout her pregnancy. It is suspected that the child may have congenital toxoplasmosis. Which immunoglobulin level measurement is most appropriate to better indicate a past infection in the new-born?
- A. Toxoplasma IgG
 - B. Toxoplasma IgE
 - C. Toxoplasma IgA
 - D. Toxoplasma IgM
- ANSWER: A
119. One morning, a female laboratory technologist was doing an ELISA test. She added the sample antigen into the microplate well and incubated the antigen in the microplate to be adsorbed onto the well surface, and then washed. She added enzyme-conjugated complementary antibodies to the target antigen, incubated and washed in order to remove any unbound antibodies. She added substrate to stop the enzyme-substrate reaction. Finally, she detected the product based on the intensity of the colour. What is the most likely type of ELISA the technologist did in this case?
- A. Indirect
 - B. Direct
 - C. Sandwich
 - D. Competitive
- ANSWER: B
120. A 46-year-old female from rural part of malaria endemic area visited the nearby health post with intermittent high fever, shivering and headache. However, the health post did not have microscope and electric power to confirm whether or not she had malaria. What is the most likely alternative to diagnose malaria in this case?
- A. PCR
 - B. RDT
 - C. ELISA
 - D. Gram staining
- ANSWER: B

121. A 10-year-old boy, who is living in a slumped, stool-contaminated area and walks barefooted complains an abdominal pain for three weeks. His mother took him to the nearby health centre. A physician took the history of the child, made an examination, and he noticed that the child was anaemic and has bloody diarrhoea. Then, the child was sent to a laboratory for stool sample examination.

What is the most likely parasite the child might acquire in this case?

- A. *Hook worm*
- B. *Garida lamblia*
- C. *Trichuris tricurua*
- D. *Ascaris lumbricoids*

ANSWER: A

122. An 8-year-old boy was clinically diagnosed with weight loss, abdominal cramps, and nausea. A laboratory technologist examined the stool using the direct wet mount method and didn't detect any parasites. The technologist wants to perform another method to increase the parasite load in the specimen preparation and make it easier to detect.

Which method would be most appropriate for the intended purpose?

- A. using polyvinyl alcohol fixative method
- B. Applying 5 percent ligolsiodin
- C. Formol ether concentration method
- D. Applying 0.85 sodium chloride solution

ANSWER: C

123. A 40-year-old man came to a clinic with complaints of abdominal pain, nausea, cough, and weakness. After a thorough examination of the patient, a physician suspected bronchopneumonia secondary to intestinal parasitic infection and requested stool examination. The laboratory personnel prepared a wet mount slide for the identification of the parasite.

What would be the most likely parasite to be investigated in this case?

- A. Adult stage of *Ascaris lumbricoides*
- B. Larval stage of *Entrobis vermicularis*
- C. Larval stage of *Strongyloides stercoralis*
- D. Egg stage of *Trichuris trichiura*

ANSWER: C

124. A graduating medical laboratory student has conducted research as partial fulfillment of a BSc degree. A stool specimen was collected from students of an elementary school found in a small district. After doing a laboratory investigation on the collected stool sample, he identified yellowish brown, oval to round eggs with a thick double shell and a single round granular central mass.

What is the most likely parasite identified with this morphological feature?

- A. Hook worm
- B. *Ascaris lumbricoides*
- C. *Entrobis vermicularis*
- D. *Strongyloides stercoralis*

ANSWER: B

125. A stool specimen collected from an 8-year-old girl was prepared by a technician, and he examined the whole field and reported that a moderate number of parasites were identified.

What would be the most likely number of parasites seen per slide?

- A. 1-3 per slide
- B. 11-20 Per slide
- C. 4-10 Per slide
- D. 21-40 Per slide

ANSWER: B

126. A public health officer working in a health station which is located in an area where infections with *Wuchereria bancrofti* and *Onchocerca volvulus* are frequently detected and the report is increasing. So, the officer decides to work on introducing prevention and control strategies for these parasitic infections in the area.
What strategy is most likely appropriate in this case?
- A. Avoiding the use of night soil as a fertilizer
 - B. Avoiding walking barefoot
 - C. Using insecticides to destruct vectors
 - D. Wash hands before a meal
- ANSWER: C
127. A researcher wants to conduct research in the area where blood and tissue nematodes are endemic. He wants to know the level of distribution of these parasites in a population.
What would be the most appropriate method of diagnosis to identify this parasite?
- A. Identify the parasite from the urine sample using the concentration technique
 - B. Identify the parasite from the thick blood film collected at the best timing
 - C. Concentrated capillary blood after haemolysing by saponin-saline
 - D. skin specimens from patients with nodules were investigated using the wet slide method
- ANSWER: C
128. A 45-year-old man came to a hospital with abdominal pain, loss of appetite, and intestinal disturbance. Geographically, he came from an area where cattle are raised and undercooked/raw beef meat is eaten. An attending physician suspects taeniasis.
What would be the most appropriate type specimen in this case?
- A. Blood specimen
 - B. Skin snap
 - C. Body fluid
 - D. Stool specimen
- ANSWER: D
129. A 17-year-old boy, who lives in a small village surrounded by a lake in the highland region of Ethiopia, was diagnosed with schistosomiasis.
What is the most likely infection the boy acquires in this scenario?
- A. Penetration of skin by miracidia
 - B. Penetration of skin by cercaria
 - C. Penetration of skin by adult worm
 - D. Penetration of skin by sporocyst
- ANSWER: B
130. A laboratory technologist, who was working in a hospital, asked for semi-concentration and semi-quantitative estimation of schistosome eggs in stool specimens by community workers of schistosomiasis epidemiology and control work.
What would be the most appropriate and efficient method to use in this case?
- A. Stool Wet mount method
 - B. The Kato–Katz technique
 - C. Giemsa stain technique
 - D. Right stain method
- ANSWER: B

131. A laboratory technologist received a dysenteric faecal specimen collected from a patient. He assembled all the materials needed for faecal examination. He also wanted to place a small amount of the specimen on the slide.
What would be the most appropriate sampling method for the identification of the parasite?
A. Take a sample from the loose part of the stool specimen
B. Take a sample from the blood and mucus part
C. Mix the specimen and take the sample
D. The sampling doesn't have an effect.
ANSWER: B
132. A water source of an elementary school was reported to be contaminated with public waste disposal pipe, and a number of children were sick and presented with foul-smelling diarrhoea, abdominal pain, bloating, and vomiting. The stool specimens collected from those children showed Giardiasis.
What is the most likely means of acquiring the infection in this case?
A. Consuming water containing cysts
B. Consuming water contaminated with trophozoites
C. Penetration of the skin by the cyst stage while bathing
D. Penetration of the skin by the trophozoite stage while bathing
ANSWER: A
133. A public health official of a zonal health department planned to perform prevention and control activities so as to reduce the transmission of *Cryptosporidium*, *Cyclospora* & *Isospora* among the community.
Which is NOT the most appropriate preventive method to be applied in this case?
A. Safe water supply
B. Sanitary disposal of faeces
C. Eradicate the intermediate host
D. Avoid use night soil as a fertilizer
ANSWER: C
134. A diaspora who came to our country was suspected of having acquired visceral leishmaniasis. But, repeated blood examinations gave a negative result. Observation of LD bodies may confirm the presence of the parasite.
What is the most likely sensitive specimen to detect the LD bodies microscopically?
A. Liver biopsy
B. Lymph node aspirates
C. Bone marrow aspiration
D. Splenic aspiration
ANSWER: D
135. A fourth-year laboratory student found a stage of a parasite which is intracellular in the reticulo-endothelial cells that was lying separately between the cells. It is measured 2–4mm and has a prominent nucleus and a rod-shaped kinetoplast. Both the nucleus and the kinetoplast stain are red, and the cytoplasm stains are pale blue.
What is the most likely parasite with this morphological feature?
A. *T. gambiense*
B. *T. rhodesiense*
C. *T. brucei*
D. *T. cruzi*
ANSWER: D

136. A nurse who served for 10 months in an emergency OPD has a history of repeated needle stick injury. She didn't travel in her life to a place which is below 2500 meters above sea level. She developed periodic fever, chills, and sweating. She suspected malaria infection and gave blood for examination.

What is the most likely mode of infection of the causative agent?

- A. By direct mode of transmission
- B. By the indirect mode of transmission
- C. By bite of a female *Anopheles* Mosquito
- D. By congenital mode of transmission

ANSWER: B

137. A 35-year-old woman was infected with Toxoplasmosis during antenatal care. A physician ordered an ultrasound examination. She showed that her foetus developed Microcephaly and hydrocephaly during the last trimester.

At what diagnostic stage would the disease most likely be transmitted to the foetus?

- A. Bradyzoites stage
- B. Tachyzoites stage
- C. Sprozoite stage
- D. Merozoite stage

ANSWER: B

138. A senior laboratory technologist, who was working in a given teaching hospital, was discussing about a Giemsa-stained microfilaria with senior medical laboratory technology students attached to his lab for practice. The microfilaria under the microscope was thick, nuclei not extending to the tip of the tail, and didn't have any sheath.

What microfilaria is most likely characterized by the absence of a sheath when stained by Giemsa solution?

- A. *B. malayi*
- B. *L. loa*
- C. *O. volvulus*
- D. *W. bancrofti*

ANSWER: C

139. A 37-year-old patient had abdominal pain and consistent watery diarrhea for the last two weeks. The patient was clinically evaluated by a physician. Finally, he was sent for stool examination. A laboratory technologist prepared all the necessary materials, including staining solutions for opportunistic intestinal parasites.

What is the most appropriate type of staining solution is use to stain the stool sample in this case?

- A. Gram stain
- B. Giemsa stain
- C. Methylene blue stain
- D. Ziehl Neelsen stain

ANSWER: D

140. A fresh graduate medical laboratory technologist was employed at a given parasitological research laboratory. He was doing identification of opportunistic parasitic protozoan parasites using modified acid-fast staining. While examining a stool sample, he found round and irregularly stained bodies containing red-staining granules, a central refractile morula-like structure, and some oocysts that were unstained and appeared glassy, wrinkled spheres. What is the most likely diagnosis in this case?
- A. Isospora species
 - B. Cryptosporidium species
 - C. Cyclospora species
 - D. Microsporidia species
- ANSWER: C
141. Water for laboratory use must be free from contaminants that could interfere with the performance of tests. The quality of purified water is graded according to the amount of biological matter as Grade 1, 2 or 3 (ISO 3696: 1987). Among these three water grades, one is recommended for the analytical work carried out in district hospital laboratories with full facilities. What water grade is the most likely recommended based on the above scenario?
- A. Grade 1
 - B. Grade 2
 - C. Grade 3
 - D. Grade 2 and 3
- ANSWER: C
142. The accuracy and precision of a pipette used in sample or standard solution dilution is critical to avoid analytical errors since it is an important aspect of quality assurance. Accordingly, a laboratory technologist has recorded different sample results for long-term monitoring and to track changes over time with a particular pipette. What pipette verification type is most likely to be used in this case?
- A. Aspiration
 - B. Validation
 - C. Calibration
 - D. Purification
- ANSWER: C
143. A fully automated spectrophotometer was placed in a clinical chemistry laboratory near the window where excessive dust can enter the laboratory. Lots of dust particles are observed anywhere in the light path, which causes any wavelength outside the band to be transmitted by the monochromator and affect the result of the analyte of interest to be measured at a unique wavelength. What is the most likely factor that would interfere with the measurement in this case?
- A. Outdated reagent
 - B. Wavelength inaccuracy
 - C. Concentration of the analyte
 - D. Radiation beyond the interest of wavelength
- ANSWER: D

144. A postgraduate student was interested in separating solute fractions based on their solubility in the solvent, the rate of diffusion, and the polarity of the solvent and solutes in the sample. By using a chromatographic technique, a thin layer of Silica gel is attached to a glass plate as a stationary phase and a liquid solvent as a mobile phase. This technique operates by moving the fractions of the sample solubilized in the solvent across the stationary phase by capillary action. Which chromatography technique is most likely to function in this case?
- A. Column Chromatography
 - B. Column Chromatography
 - C. Thin-Layer Chromatography
 - D. Partition column chromatography
- ANSWER: C
145. A newly assigned laboratory professional wants to prepare a reagent solution for clinical chemistry tests. What is the most likely apparatus used for the measurement of liquid volume?
- A. Beaker
 - B. Burettes
 - C. Flasks
 - D. Reagent bottle
- ANSWER: B
146. A laboratory technologist, who was working in a district health centre where an autoclave was used for sterilization. He wanted to sterilize culture media used for bacteriological examination. What is the most likely care the technologist has to take during the process?
- A. Make sure that the water touches the basket
 - B. Wipe the inside of the autoclave daily or whenever spillages occur
 - C. Heat the autoclave too quickly to bring up the pressure once the outlet valve is closed
 - D. Leave the autoclave to cool for too long
- ANSWER: B
147. A 40-year-old male client brought a request form and urine specimen to the main laboratory of a hospital at 5:00 am (local time). The laboratory personnel hesitated to receive the request of the specimen because there was a considerable delay. What could be the most likely reason for the technician to reject the sample?
- A. Organism present would be decreased within that urine.
 - B. Gives a false result of a possible infection
 - C. The cell count could be increased
 - D. The sample could have insufficient volume
- ANSWER: B
148. A laboratory technician working in the microbiology department wanted to sterilize culture media for another laboratory work. What is the most appropriate sterilization method needed in this case?
- A. Boiling
 - B. Disinfection
 - C. Autoclave
 - D. Dry hot oven
- ANSWER: C

149. Medical laboratory personnel were exposed to many dangers, not only from infected material but also from dangerous chemicals and apparatus that they use routinely. Precautions must be taken by all members of the staff for the safety of all concerned.
Which safety precaution measure is NOT most appropriate to minimize laboratory accidents?
A. Bottles must be clearly labelled to show their contents.
B. Discard broken glasses into the waste paper basket.
C. Examine glassware before use
D. Identify hazardous chemicals within the laboratory
ANSWER: B
150. A laboratory technologist, who was working in a medium clinic laboratory, wanted to perform a urine microscopic examination. The technologist is required to use equipment to hasten the deposition of substances suspended in the urine specimen.
What could be the most likely laboratory equipment used in this case?
A. Autoclave
B. Centrifuge
C. Distiller
D. Microscope
ANSWER: B
151. A laboratory technologist, who was hired in a new histopathology laboratory, started organizing the laboratory and making requests for purchases. The technologist wanted to choose among the various fixation techniques.
What is the most likely fixative technique used in tissue processing in this case?
A. Heat fixation,
B. Microwave fixation
C. Freeze-drying
D. Chemical fixation
ANSWER: D
152. A comprehensive specialized hospital decides to establish a new histopathology laboratory. A purchasing team has been organized to purchase reagents, materials, and equipment necessary for the establishment of the laboratory. The team wants to purchase the most common infiltration and embedding medium so as to ensure availability and easiness in application.
What the most likely recommended medium to be purchased in this case?
A. Resin
B. Celloidin
C. Paraffin wax
D. Gelatin
ANSWER: C
153. In the hospital's medical laboratory department, a laboratory quality officer reviewed the Standard Operation Procedure (SOP) of the pre-analytical phase of quality assurance. Unfortunately, the possible source of error that should be included under the pre-analytical phases was missed. Therefore, he planned to rewrite the SOP for the pre-analytical phase.
What is the most likely source of error that would be included in the SOP?
A. Use of faulty equipment, improper use of equipment
B. Incorrect reagent preparation and storage
C. Incomplete laboratory request forms
D. Inaccurate calculations, computation or transcription
ANSWER: C

154. A medical laboratory scientist received a blood specimen from a 44-year-old male, which was collected by an experienced nurse for renal function test analysis. The laboratory reception received the specimen as per the standard operational procedure (SOP) and sent it to a clinical chemistry department. Finally, the issued laboratory result was not in line with the previous day result. The Laboratory head investigated the issue and concluded as if there was a possible problem at the analytical stage and decided to repeat the RFT test from the leftover sample. What is the most likely way of preventing the source of error in this case?
- A. Transportation of the sample to the laboratory
 - B. Incubation time of the sample with the reagent
 - C. How to interpret the laboratory result
 - D. How to collect a sample from the patient
- ANSWER: B
155. A biotechnology department student, who developed a new reagent, which was used to analyse X analyte from the biological fluids collected at the Medical Laboratory department. Among the parameters, a student evaluated the performance of the reagent by doing the linearity check experiment. What action needs to be considered during this test?
- A. The lowest and the highest concentrations of analyte X to be tested by new reagent
 - B. The analyte X to be tested by the new reagent should be diluted by the appropriate diluent
 - C. Testing of analyte X Standard solutions at different concentrations is enough for the decision
 - D. Pearson's correlation coefficient (r) slope close to "0" shows a better accuracy result
- ANSWER: A
156. A hospital employed a fresh medical laboratory scientist, prepared refreshment training on quality laboratory services. During their training, the qualities of good standard operation procedures (SOP) were one of the agenda to be addressed by the trainer. What would be the most likely characteristic of good SOPs to be addressed by the trainer?
- A. Easily understood by new personnel
 - B. Reviewed and approved by the quality officer
 - C. Brief, and mainly focused on the analytical steps
 - D. Must keep confidential at the quality officer office
- ANSWER: A
157. In a newly established hospital, a medical laboratory department head recruited few laboratory staff. The head arranged a meeting to discuss on preparation and implementation of the standard operation procedure (SOP). What point is the most likely to be included by the laboratory head for the recruited staff?
- A. Rely solely on the manufacturer's product inserts for SOPs
 - B. The SOP and job aid are synonymous terms and have similarities when prepared
 - C. The SOP should contain general instructions, rather than step-by-step details
 - D. The SOP should refer to any relevant procedures that may be written separately
- ANSWER: D

158. A laboratory technologist, who was working in a hospital's medical laboratory department of clinical chemistry unit, runs quality control tests, and interprets results before running patient samples. The technologist observed that the glucose normal control material was moving in one direction towards +3SD for the last six days, and appears to be heading toward an out-of-control value. He noticed that the quality control has failed and started to assess the reason for the failure. What is the most likely reason for the failure of quality control in this scenario?

- A. Random error
- B. Shift
- C. Trend
- D. Bias

ANSWER: C

159. A hospital management committee requested the medical laboratory department to participate in the external quality assurance (EQA) program. The management team advised a laboratory department to participate in EQA program that can give a true picture of a laboratory's overall performance.

What is the most likely EQA program used in this case?

- A. A. On-site evaluation
- B. B. Blind Rechecking
- C. C. Retesting
- D. D. Proficiency testing

ANSWER: A

160. A senior class of medical laboratory students had a laboratory attachment in a referral hospital of clinical chemistry laboratory. During their visit, a laboratory quality officer explained how they draw the quality control (QC) chart and record the quality control result.

What is the most likely need to be registered in association with the signature and date in this case?

- A. The hospital laboratory medical laboratory head
- B. The hospital laboratory clinical chemistry unit head
- C. Laboratory professional who performs QC tests
- D. The hospital laboratory quality officer

ANSWER: C

161. Senior class medical laboratory Science students had a visit to a Biotechnology Company that develops and manufactures different medical laboratory reagents and diagnostic kits. During the students' visit, the head of the test kit development unit refreshes students about the definitions of sensitivity and specificity terms.

What is the most likely described in the explanation of the unit head regarding specificity?

- A. The fractions of those with a specific disease that the assay correctly predicts
- B. The fractions of those without a specific disease that the assay correctly predicts
- C. The assay predicts the proportion of subjects with a negative test who do not have the disease
- D. The assay predicts the proportion of subjects with a positive test who have the disease

ANSWER: B

162. A hospital management team noticed that there was a client satisfaction problem at the medical laboratory department. Thus, to solve the issue, the team organized as the hospital's medical laboratory head and laboratory quality officer to discuss the matter thoroughly.

What is the most likely strategy for managing the client satisfaction problem in this case?

- A. Assigning a spy technologist to follow each laboratory professional's activity
- B. Prepare an occurrence log book to record errors and any actions taken
- C. Keep the occurrence report and management actions secret from the management team
- D. Make monitoring of occurrences as the sole responsibility of the laboratory quality officer

ANSWER: B

163. A regional health bureau manager decided to implement a project that reduces mortality related to chronic disease in a zone. The project team conducted a short survey and identified that hypertension, diabetic mellitus, and cancer were the main causes of mortality in the community. To select the problem, the team assessed the consequent suffering and disability on each type of disease.
What is the most likely problem prioritization criteria applied by the team?
- A. Relevance
 - B. Feasibility
 - C. Political acceptability
 - D. Avoidance of duplication
- ANSWER: A
164. Health science researchers want to conduct a study on the presence of malaria disease in a *Woreda's* community. They also want to identify the presence or absence of the exposure variables at the same point in time.
What is the most likely appropriate study design in this case?
- A. Cross-sectional
 - B. Case-control
 - C. Experimental
 - D. Cohort
- ANSWER: A
165. A team of researchers is planning to conduct a study to determine the prevalence of hypertension among adults aged 40 and above in a rural community. In a rural kebele, there are 25 to 30 villages where each village comprises of 30 households. The villages are far distant from each other. A researcher decided to include all adult population in 10 villages.
What is the most likely random sampling technique to select the study participants in this case?
- A. Stratified
 - B. Simple
 - C. Cluster
 - D. Multi-stage
- ANSWER: C
166. A team of researchers was interested in studying the prevalence of diabetes mellitus among adult population in rural villages of Ethiopia. The researchers included adults of thirty years and more to be part of the study.
What is the most likely study unit in this study?
- A. All residents in the rural villages
 - B. Individual households in the rural village
 - C. All individuals with diabetic diagnoses
 - D. All adults aged 30 years and above
- ANSWER: D
167. A research team was analysing data collected from a survey on dietary habits and chronic health risk factors in a sample of adults in a community. Upon data review, they noticed inconsistencies and missing values in some responses.
What is the most likely data quality improvement technique to be used in this study?
- A. Implementing data validation check
 - B. Hiring additional staff to manually clean
 - C. Removing incomplete data from data set
 - D. Conducting a survey with the same participant
- ANSWER: A

168. A research team conducted a study to determine the association between sleep duration and mental health outcomes among adolescents in a community. A team collected self-reported data and started data cleaning for analysis. However, the researchers encountered inconsistencies of data with some unrealistic data values.

What is the most likely data quality improvement technique to be used in this study?

- A. Implementing data validation checks
- B. Excluding the participant
- C. Conducting follow-up review of the questionnaire
- D. Using a sleep monitoring device

ANSWER: A

169. A graduate medical laboratory student interested in assessing the CD4 count of patients living with HIV, attending the ART clinic in a district, and referral hospitals. The student collected the data and checked for normality assumption and found it to be normally distributed.

What is the most appropriate measure of central tendency to be used in this case?

- A. Mean
- B. Median
- C. Mode
- D. Proportion

ANSWER: A

170. A research team conducted a study to compare two different exercise programs in reducing body weight among obese individuals. The team collected data on participants' weight at baseline, 6- and 12-weeks following the exercise program. A team wants to present the findings in a clear and informative manner for publication in a scientific journal.

What is the most appropriate data presentation method to be used in this study?

- A. Bar chart
- B. Line graph
- C. Pie chart
- D. Table listing

ANSWER: B

171. A research team conducted a study to investigate the association between exposure to air pollution and the development of respiratory disease in a city. The researcher collected data from 1000 individuals, categorized them into those exposed to high levels of air pollution and those exposed to low levels of air pollution. After following the participants for five years, the researcher found that among those exposed to high levels of air pollution, 100 developed respiratory diseases, while among those exposed to low levels of air pollution, 50 developed respiratory diseases.

What is the most appropriate measure of association to be used in this study?

- A. Rate ratio
- B. Odds Ratio
- C. Relative Risk
- D. Hazard ratio

ANSWER: C

172. A researcher is interested to study on the change in CD4 count over time. The researcher wants to inform what CD4 count mean to the readers.

In which part of the most appropriate proposal section should it be described?

- A. Introduction
- B. Statement of the problem
- C. Literature review
- D. Justification of the study

ANSWER: A

173. An executive manager of a diagnostic centre has aimed to expand organizational services by incorporating new diagnostic examinations like MRI and CT scans. The manager prepared lists of activities that need to be accomplished in the coming year, such as establishing a team of experts, individuals responsible for tasks, allocating necessary resources, and setting a timeline to accomplish different tasks.

What is the most likely management function the managers execute in this case?

- A. Planning
- B. Directing
- C. Organizing
- D. Controlling

ANSWER: A

174. A diagnostic centre wants to develop a plan for the upcoming year. The centre organizes a team of experts responsible for conducting a survey on the size, composition, and distribution of the target population in the catchment area. The team also reviewed documents to understand the population characteristics, infrastructure, and health needs of the community before endorsing specific actions.

What is the most likely planning step carried out by the team in this case?

- A. Situational analysis
- B. Action plan development
- C. Selecting priority problems
- D. Setting objectives and targets

ANSWER: A

175. A private health science college has planned to open a master 's-level program in a clinical laboratory track. The college has assigned a team of staff to conduct a survey on the organization's capacity and readiness to open the program. At the end of the survey, the team identifies the availability of committed staff, a good laboratory simulation setup, and adequate resources to open the program.

What is the most likely component of SWOT analysis that is described in this case?

- A. Strength
- B. Weakness
- C. Opportunity
- D. Threat

ANSWER: A

176. A medical laboratory unit head delegated one of his staff to compile a monthly infection control report. For the first month, the delegated staff member prepared the report but forgot to submit it to the unit leader. For the next six months, the delegated staff member totally ignored the task and focused on routine activities only. Suddenly, the unit leader recalled that the delegated staff member never sent him the monthly report and shouted at him.

What is the most likely effective delegation principle that is missed by the leader?

- A. Assess results
- B. Progress follow-up
- C. Provision of authority
- D. Control and coordinate

ANSWER: B

177. A head of the medical diagnostic centre manager is preparing a human resource plan for the coming year. As a method of forecasting human resource needs, he developed a questionnaire and anonymously surveyed the opinions of experts. He analysed the opinion and sent it back to the experts for further evaluation on the appropriateness of the forecast.
What is the most likely expert opinion forecasting method is applied by the manager?
- A. Trend projection
 - B. Delphi technique
 - C. Nominal technique
 - D. Forecasting demand
- ANSWER: B
178. A medical laboratory unit head assigned a new medical laboratory unit staff member to work under the supervision of an experienced staff member. The intention of the unit head is to help the new staff member learn and handle the activities under supervision.
What is the most likely training method the head applied in this case?
- A. Tutorials
 - B. Job rotation
 - C. Mentoring
 - D. Off-the-job training
- ANSWER: C
179. A head of a medical diagnostic centre recognizes a conflicting situation among the staff members. He is not assertive in pursuing the interest of the diagnostic centre and ignores the conflicting situation to be resolved on its own time.
What is the most likely conflict management strategy followed by the head?
- A. Avoidance
 - B. Compromise
 - C. Collaboration
 - D. Accommodation
- ANSWER: A
180. An international medical laboratory centre wants to purchase materials and supplies for its branch service for the next month. The centre purchased a laboratory supply from the vendor that is related to it.
What is the most likely procurement method followed by a laboratory centre in this case?
- A. Open tender
 - B. Restricted tender
 - C. Direct procurement
 - D. Competitive bidding
- ANSWER: C
181. A store manager of a primary hospital issued materials by preparing and keeping a document that consisted of date, type of material, and amount of the material to the departments. The responsible and issuing personnel's names and signatures is well recorded.
What is the most likely paperwork prepared by the store manager in this case?
- A. Issue Voucher
 - B. Ledger Record
 - C. Inventory recorded
 - D. Distribution record
- ANSWER: A

182. A manager of a general hospital has been leading the organization for nine years. He is just a figurehead and does not give any direction to the subordinates. He delegates the authority to subordinates so that they must plan, motivate, control and be responsible for their actions. What is most likely style of leadership described in this case?
- A. Autocratic
 - B. Democratic
 - C. Participative
 - D. Laissez-faire
- ANSWER: D
183. In a district hospital, the request of blood glucose concentration of diabetic patients on follow-up decreased after a newly assigned medical laboratory technologist. In order to understand the laboratory technologist's rapport with clients, the laboratory quality officers need to actively seek information from clients. What is most likely important to evaluate the ability of the technologist in establishing rapport with the clients?
- A. Conducting a Client satisfaction survey
 - B. Conducting External quality assessment
 - C. Developing a standard operating procedure
 - D. Preparing quality management exam for the technologist
- ANSWER: A
184. A female medical geneticist is interested in collecting positive blood samples for *Plasmodium falciparum* to study the gene that is responsible for anti-malarial drug resistance. Before any data collection, she communicated with the head of the laboratory to appropriately store the positive samples and record the patient's name on the registration book that matches the label on patient's sample. However, when she revisited a hospital laboratory to assemble the samples, there was no patient name on the registration for certain samples. What is the most likely reason in this case?
- A. Lack of instructions
 - B. Lack of equipment
 - C. Lack of good recording
 - D. Optimum temperature
- ANSWER: C
185. A laboratory technologist was examining the complete blood count. A mother who exhibits palmer pallor and pale conjunctiva obtained normal haemoglobin laboratory findings. When the problem was investigated back, there was a machine calibration problem that the technologist needed to check before reporting. As a result, the staff accused him of his practice. What ethical principle is most likely breached by the fresh graduate?
- A. Autonomy
 - B. Beneficence
 - C. Nonmaleficence
 - D. Veracity
- ANSWER: C

186. An attendant comes to the laboratory department to take the lab results for his relatives. A junior medical laboratory technologist gives the result and warns the relative to protect the result from anybody and to keep it a secret. Then, the relative asked the technologists the reason for this action. He replied that they have an infectious disease in the blood.
What ethical principle is most likely justifying the malpractice committed by the technologist?
- A. Autonomy
 - B. Beneficence
 - C. Confidentiality
 - D. Informed consent
- ANSWER: C
187. A medical laboratory technologist, who was working in a referral hospital, always pretends to know everything. He doesn't want to consult or discuss with a workmate. He fails to acknowledge the limits of his professional knowledge and competence.
What is the most likely professional duty that needs to be treated for the technologist?
- A. Duty to self
 - B. Duty to the patient
 - C. Duty to society
 - D. Duty to profession
- ANSWER: A
188. A medical laboratory technologist always seeks training and reads high-stakes clinical trials to update day-to-day medical laboratory practice.
What principle of professionalism is most likely addressed in this case?
- A. Altruism
 - B. Humanism
 - C. Accountability
 - D. Clinical excellence
- ANSWER: D
189. A commander- in -chief of military personnel visited a laboratory department for lab investigations. Accidentally, the laboratory science personnel called the client without the prefix 'commander' to give him the result. The client became hostile and reported to the hospital administration, and the technologist was accused of not working as per the national CRC protocol.
What is the most likely component of dignity breached by a laboratory technologist?
- A. Dignity of humanity
 - B. Dignity of personal identity
 - C. Dignity of merit
 - D. Dignity of moral status
- ANSWER: C
190. A medical laboratory science professional who is working in a health center is assigned to the outreach school health service team. The team delivered health education for high school students on the effects of peer influence on risky sexual practices and its results of acquiring sexually transmitted diseases.
What factors of health behaviour is most likely explained in this case?
- A. Enabling factors
 - B. Aggravating factors
 - C. Predisposing factors
 - D. Reinforcing factors
- ANSWER: D

191. Infectious Prevention and Waste Management Committee conducted a facility inspection in a general hospital. The committee found improperly disposed human tissues, stained body fluid, waste materials, and human body parts in the compounds.
What is the most likely type of waste generated in this case?
- A. Pathological wastes
 - B. Human specimens and blood products
 - C. Cultures and stocks of infectious agents
 - D. Shares used in diagnosis or inoculation
- ANSWER: A
192. Health professionals were exposed to biohazards and developed a serious infectious disease in a scientific laboratory. As a result, professionals received instructions on how to handle highly toxic materials that can be released into the air by using a mechanical handling device or a closed glove system that can be operated from the outside.
What is the most likely workplace safety practice explained in this case?
- A. Isolation
 - B. Enclosure
 - C. Ventilation
 - D. Elimination
- ANSWER: B
193. A regional health department established a team to investigate the cholera epidemic based on the reports from the woreda health office. The team confirmed the diagnosis conducted by clinical laboratory test, and also the existence of an epidemic by comparison with previous incidence of the disease in the same catchment population.
What is the most likely next step to be taken in this scenario?
- A. Verify the diagnosis
 - B. Formulate hypotheses
 - C. Describe the epidemic
 - D. Search for additional cases
- ANSWER: C
194. A 60-year-old male patient came to a general hospital with a complaint of bleeding during defecation. A laboratory test and other diagnostic procedures proved colon-rectal cancer. Health professionals provided rehabilitation care to the patient during the stage of the disease.
What level of disease prevention is most likely explained in this case?
- A. Primary
 - B. Tertiary
 - C. Primordial
 - D. Secondary
- ANSWER: B
195. A traveller wanted to take malaria prevention methods while travelling to malaria endemic area in a country. The traveller consulted a health professional on the recommended malaria chemoprophylaxis.
What is the most likely outbreak control measures explained in this case?
- A. Avoiding exposure.
 - B. Destroying the agent
 - C. Reducing the reservoir host
 - D. Reducing host susceptibility
- ANSWER: D

196. An environmental health professional wanted to recommend a technique that used to dehydrate food sources in more safer and sanitized manner to a rural village of the woreda community.
What is the most likely technique of food preservation used in this case?
- A. Drying
 - B. Freezing
 - C. Blanching
 - D. Fermentation
- ANSWER: A
197. In a region, food was served for a large number of people in a refugee camp. After consuming the food, a higher number of patients arrived at the nearby health facility with a disease caused by contaminated food with chemical poisoning. A laboratory finding indicated fungal toxins in the sample.
What is the most likely cause of the disease in this case?
- A. Botulism
 - B. Aflatoxin
 - C. Perfringens
 - D. Neurolathyrism
- ANSWER: B
198. A public health microbiologist wants to conduct microbial analysis of a food sample. In his method, homogenised and dilutions of food samples are prepared. Three serial aliquots or dilutions are then planted into 9 or 15 tubes of appropriate medium for the three or five-tube method, respectively. Numbers of organisms in the original sample are determined by the use of standard tables.
What is the most likely method used in this case?
- A. Most probable number
 - B. Howard Mold counts
 - C. Microscopic colony count
 - D. Agar droplets
- ANSWER: A
199. A 54-year-old patient who has a history of cattle contact develops a high, irregular fever with shivering, sweating, and an increased pulse rate. Blood film examination is requested. The result of thick blood film examination shows *Trypanosoma brucei rhodesiense* (*T. b. rhodesiense*).
What type of fly species is most likely a vector for the parasite indicated in this case?
- A. Sandy flies
 - B. House flies
 - C. Tsetse flies
 - D. Black flies
- ANSWER: C
200. A man living around a fast-flowing River has visited a health facility with clinical manifestations of intense skin irritation, raised papules on the skin, and subsequently alteration in the pigmentation of the skin.
Which vector is most likely responsible for the patient's problem?
- A. Ticks
 - B. Sand flies
 - C. Black flies
 - D. Female anopheles mosquito
- ANSWER: C